

Exercise Sheet 2: Random Variables and Probability Distributions

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Issued: 6 September 2026 • Due: 11 September 2026

About this sheet. There are fifteen questions in this exercise sheet. When you write a distribution function, pay attention to the end-points of the intervals. For transformation questions, first write the event in terms of the original random variable.

Source. The sheet is based mainly on Chapter 2 of Rohatgi and Saleh, especially Problems 2.2–2.5, with the wording simplified and some questions modified or added.

Part A. Random variables and distribution functions

1. Two fair dice are rolled. Let X be the sum of the two face values and let Y be their absolute difference.

- (a) List the possible values of X and of Y .
- (b) Write $\{X \leq 4\}$ and $\{Y = 1\}$ explicitly as subsets of

$$\Omega = \{(i, j) : 1 \leq i, j \leq 6\}.$$

2. Let X be a real-valued random variable on $(\Omega, \mathcal{S})$.

- (a) Show that X^2 is a random variable by expressing the event $\{X^2 \leq t\}$ in terms of X .
- (b) Construct a counterexample to the converse statement.

You may use the criterion that Z is a random variable if $\{Z \leq t\} \in \mathcal{S}$ for every $t \in \mathbb{R}$.

3. Return to the random variables X and Y in Question 1.

- (a) Find the distribution function $F_X(x) = \mathbb{P}(X \leq x)$ for every real x .
- (b) Find the distribution function $F_Y(y) = \mathbb{P}(Y \leq y)$ for every real y .
- (c) Sketch both graphs and determine their points of discontinuity.

4. Decide which of the following functions are distribution functions. If a function is not a distribution function, state exactly which required property fails.

- (a) $F(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < 1, \\ 1, & x \geq 1; \end{cases}$
- (b) $F(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1} x, -\infty < x < \infty;$
- (c) $F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-x}, & x \geq 0; \end{cases}$
- (d) $F(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < \frac{1}{2}, \\ 1 - x, & \frac{1}{2} \leq x < 1, \\ 1, & x \geq 1; \end{cases}$
- (e) $F(x) = \begin{cases} 0, & x \leq 0, \\ \frac{1}{2}, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases}$

Part B. Discrete and continuous random variables

5. Let $0 < p < 1$ and define

$$\mathbb{P}(X = k) = p(1 - p)^k, \quad k = 0, 1, 2, \dots$$

- (a) Verify that this is a probability mass function.
- (b) Find the distribution function of X .
- (c) For integers $0 \leq n \leq N$, find $\mathbb{P}(n \leq X \leq N)$.
- (d) Find $\mathbb{P}(X \geq n)$.

6. For each of the following, decide whether the given formula can be a probability mass function on $\{0, 1, 2, \dots\}$. If a constant is present, determine it when possible.

- (a) $p_k = K 2^{-k}$;
- (b) $p_k = \frac{K}{(k+1)(k+2)}$;
- (c) $p_k = \frac{K}{k+1}$.

7. Let $\theta > 0$ and

$$f(x) = \begin{cases} \theta e^{-\theta x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

- (a) Verify that f is a probability density function.
- (b) Find the distribution function F .
- (c) Find $\mathbb{P}(X > s)$ for $s > 0$.

8. Suppose

$$f(x) = \begin{cases} kx(1-x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the value of k for which f is a probability density function.
- (b) Find the distribution function F .
- (c) Find $\mathbb{P}(X > 0.3)$.

9. Consider

$$f(x) = \frac{1}{2}e^{-|x|}, \quad -\infty < x < \infty.$$

- (a) Verify that f is a probability density function.
- (b) Find the distribution function F separately for $x < 0$ and $x \geq 0$.
- (c) Find $\mathbb{P}(|X| \leq 1)$.

10. Let X have distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{2}, & x = 0, \\ \frac{1}{2} + \frac{x}{2}, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases}$$

- (a) Find $\mathbb{P}(X = 0)$.
- (b) Find $\mathbb{P}(0 < X \leq 0.6)$.
- (c) Find $\mathbb{P}(X = 0.4)$.
- (d) Is X discrete, continuous in the sense of having a density, or neither? Explain.

Part C. Functions of a random variable

Reminder. If $Y = g(X)$, then for every real y ,

$$\mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \in g^{-1}((-\infty, y])).$$

This is a useful starting point when g is not one-to-one.

11. Let X have probability mass function

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n,$$

where $0 \leq p \leq 1$. Find the probability mass function of each random variable:

(a) $Y = 2X + 1$;

(b) $Z = X^2$.

Clearly state the possible values of Y and Z .

12. Let X be a positive continuous random variable with density f , and define

$$U = \frac{X}{1+X}.$$

(a) Find the possible values of U .

(b) Find the distribution function of U in terms of the distribution function F of X .

(c) Hence find the density of U in terms of f .

13. Let X have density

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty,$$

and let $Y = X^2$.

(a) For $y > 0$, write the event $\{Y \leq y\}$ as an event involving X .

(b) Find the distribution function of Y in terms of the distribution function F of X .

(c) Differentiate to show that Y has density

$$h(y) = \frac{1}{\sqrt{2\pi y}} e^{-y/2}, \quad y > 0,$$

and $h(y) = 0$ for $y \leq 0$.

(d) Explain where the two contributions corresponding to $+\sqrt{y}$ and $-\sqrt{y}$ enter the calculation.

Part D. Further questions

14. A random angle and the arcsine density. Let Θ be uniformly distributed on $[0, 2\pi)$, and define

$$Y = \cos \Theta.$$

(a) What are the possible values of Y ?

(b) For $-1 < y < 1$, describe the event $\{\cos \Theta \leq y\}$ as an interval or a union of intervals in $[0, 2\pi)$.

(c) Show that the distribution function of Y is

$$F_Y(y) = \begin{cases} 0, & y < -1, \\ 1 - \frac{\arccos y}{\pi}, & -1 \leq y < 1, \\ 1, & y \geq 1. \end{cases}$$

(d) Differentiate to show that Y has density

$$f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1,$$

and $f_Y(y) = 0$ otherwise.

(e) The density tends to ∞ as $y \rightarrow \pm 1$. Why does this not prevent it from being a probability density?

15. Same distribution, different random variables. Choose ω uniformly from $[0, 1]$, and define

$$X(\omega) = \omega, \quad Y(\omega) = 1 - \omega.$$

(a) Find the distribution function of X .

(b) Find the distribution function of Y and show that X and Y have the same distribution.

(c) Find $\mathbb{P}(X = Y)$.

(d) Show that $X + Y = 1$ for every outcome ω . Explain in words what this example shows about the difference between a random variable and its distribution.

16. Same squares, different distributions. Construct two nonconstant random variables X and Y which do not have the same distribution, but for which X^2 and Y^2 have the same distribution. Verify both claims.

How to present your work. State the support of every PMF or PDF you obtain. When checking a proposed distribution, do not check only nonnegativity. Also check the total mass. For a distribution function, check that it is nondecreasing and right-continuous, and check the limits at $\pm\infty$. For transformations, show the event calculation before differentiating.